

CS & IT ENGINEERING



Computer Network - 2

Framing & Error Control

Lecture No. - 04

By - Abhishek Sir



ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW





Recap of Previous Lecture



Topic

Two-dimension (2D) Parity ✓

Topic

Hamming Code →



Topics to be Covered



Topic

Hamming Code

Topic

Hamming Distance

Topic

Optimum Packet Size

CS-05 ✓
CS-14

[MCQ]

[GATE-IT-2008] [2 Mark]



#Q. Data transmitted on a link uses the following 2D parity scheme for error detection :

Each sequence of 28 bits is arranged in a 4 x 7 matrix (rows r_0 through r_3 , and columns d_7 through d_1) and is padded with a column d_0 and row r_4 of parity bits computed using the Even parity scheme. Each bit of column d_0 (respectively, row r_4) gives the parity of the corresponding row (respectively, column). These 40 bits are transmitted over the data link.

The table shows data received by a receiver and has n corrupted bits. What is the minimum possible value of n ?

- A 1
- B 2
- ☒ C 3
- D 4

Ans: C

	d_7	d_6	d_5	d_4	d_3	d_2	d_1	d_0
r_0	0	1	0	1	0	0	1	1
r_1	1	1	0	0	1	1	1	0
r_2	0	0	0	1	0	1	0	0
r_3	0	1	1	0	1	0	1	0
r_4	1	1	0	0	0	1	1	0



Topic : Two-dimension (2D) Parity



Minimum number of bits corrupted in the Block = $\text{Max}[1, 3] = 3$

= Maximum [total number of row wise parity unbalanced,
total number of column wise parity unbalanced]

AT Recv:-
Even Parity

	d_7	d_6	d_5	d_4	d_3	d_2	d_1	d_0	
r_0	0	1	0	1	0	0	1	1	✓
r_1	1	1	0	0	1	1	1	0	←
r_2	0	0	0	1	0	1	0	0	✓
r_3	0	1	1	0	1	0	1	0	✓
r_4	1	1	0	0	0	1	1	0	✓
	✓	✓		✓	✓		✓		



Topic : Hamming Code



→ Single bit error-correcting code

→ Both transmitter and receiver must agree on same parity
[either "Even Parity" or "Odd Parity"]

* No. of parity bits is depend on no. of data bits



Topic : Hamming Code



→ Parity bit (R_i) placed at position $= 2^i$
[where $i = 0, 1, 2, 3 \dots$]

d_6	d_5	d_4	R_3	d_3	d_2	d_1	R_2	d_0	R_1	R_0
11	10	9	8	7	6	5	4	3	2	1

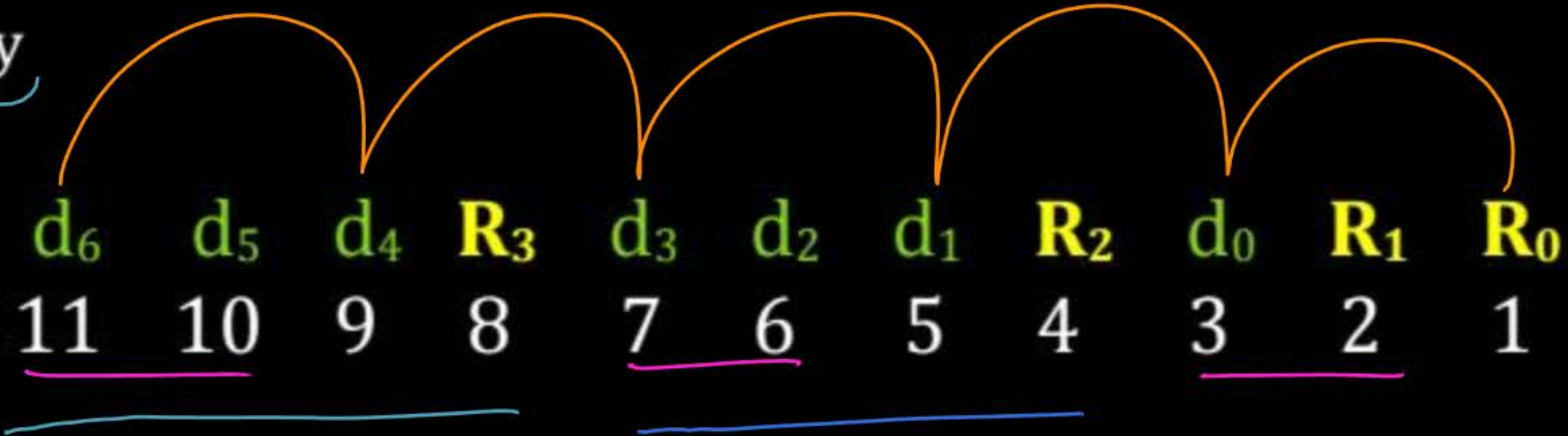
→ First parity bit placed at least significant bit position.



Topic : Hamming Code



→ Even Parity



$$R_0 = d_0 \oplus d_1 \oplus d_3 \oplus d_4 \oplus d_6$$

$$R_1 = d_0 \oplus d_2 \oplus d_3 \oplus d_5 \oplus d_6$$

$$R_2 = d_1 \oplus d_2 \oplus d_3$$

$$R_3 = d_4 \oplus d_5 \oplus d_6$$

$$[1 = 3, 5, 7, 9, 11]$$

$$[2 = 3, 6, 7, 10, 11]$$

$$[4 = 5, 6, 7]$$

$$[8 = 9, 10, 11]$$

Example 3 :-

#Q. Consider hamming code error control technique which uses 'even parity' and first parity bit placed at least significant bit position. The seven bits data '1101101' (placed from most significant bit to least significant bit position) encoded as :

Solution 3 :-

(Even Parity)

Data $\rightarrow$ 1101101



$$\begin{aligned} R_0 &= 1 \\ R_1 &= 1 \\ R_2 &= 0 \\ R_3 &= 0 \end{aligned}$$

Solution 3 :- (Even Parity)

<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>1</u>
11	10	9	8	7	6	5	4	3	2	1

Ans : 11001100111



Topic : Hamming Code



<u>d₇</u>	d ₆	d ₅	d ₄	R₃	d ₃	d ₂	d ₁	R₂	d ₀	R₁	R₀
12	11	10	9	8	7	6	5	4	3	2	1

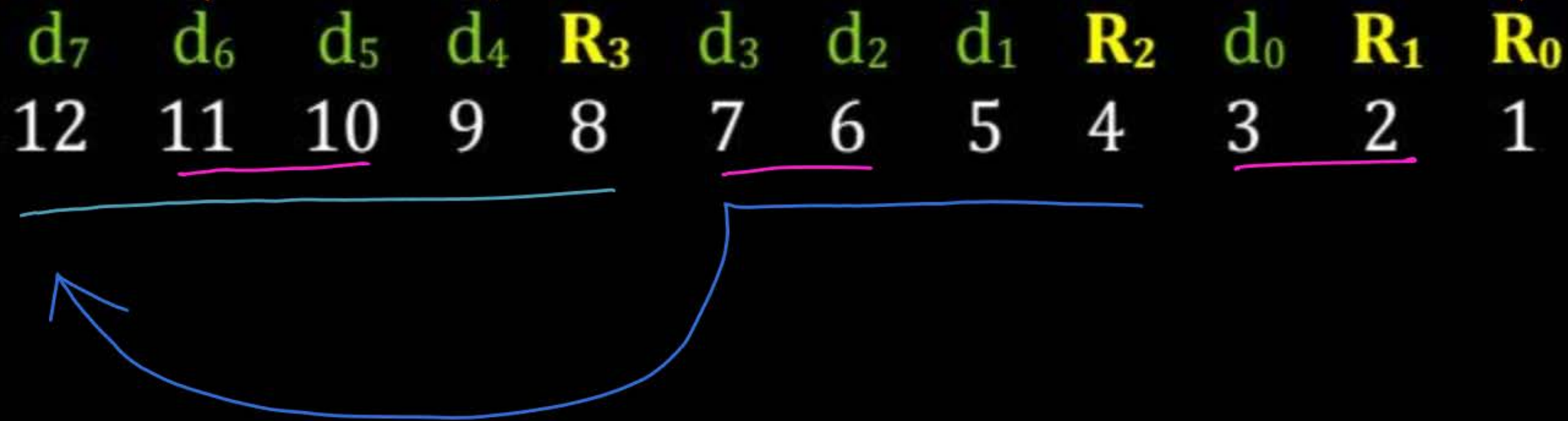
$4^{\infty} +$



Topic : Hamming Code



→ Even Parity



$$\begin{aligned} R_0 &= d_0 \oplus d_1 \oplus d_3 \oplus d_4 \oplus d_6 \\ R_1 &= d_0 \oplus d_2 \oplus d_3 \oplus d_5 \oplus d_6 \\ R_2 &= d_1 \oplus d_2 \oplus d_3 \oplus d_7 \\ R_3 &= d_4 \oplus d_5 \oplus d_6 \oplus d_7 \end{aligned}$$

$$\begin{aligned} [1 &= 3, 5, 7, 9, 11] \\ [2 &= 3, 6, 7, 10, 11] \\ [4 &= 5, 6, 7, 12] \\ [8 &= 9, 10, 11, 12] \end{aligned}$$

Example 4 :-

#Q. Consider hamming code error control technique which uses 'even parity' and first parity bit placed at least significant bit position. The ~~seven~~ ^{Eight} bits data '10101101' (placed from most significant bit to least significant bit position) encoded as :

Solution 4 :- (Even Parity)

Data $\rightarrow$ 10101101



$$\begin{aligned} R_0 &= 0 \\ R_1 &= 0 \\ R_2 &= 1 \\ R_3 &= 0 \end{aligned}$$

Solution 4 :- (Even Parity)

<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
12	11	10	9	8	7	6	5	4	3	2	1

Ans : 101001101100

←



Topic : Hamming Code



At receiver : (Even Parity)

CASE I : No any error

Transmitted Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

Received Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
12	11	10	9	8	7	6	5	4	3	2	1

P_0 ✓
 P_1 ✓
 P_2 ✓
 P_3 ✓



Topic : Hamming Code



if receiver finds all the parity bits are balanced

then receiver concluded "No any error detected" (Accept)

else

receiver concluded "Error detected" (Error correction)



Topic : Hamming Code



At receiver : (Even Parity)

CASE I : No any error

<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
12	11	10	9	8	7	6	5	4	3	2	1

Receiver Concluded : “No any error detected” (Accept the data)
[Receiver finds all parity bits are balanced]



Topic : Hamming Code

$$1+2+4=7$$



At receiver : (Even Parity)

CASE II : One-bit error

Transmitted Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

Received Codeword = 1 0 1 0 0 0 1 0 1 1 0 0

<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
12	11	10	9	8	7	6	5	4	3	2	1

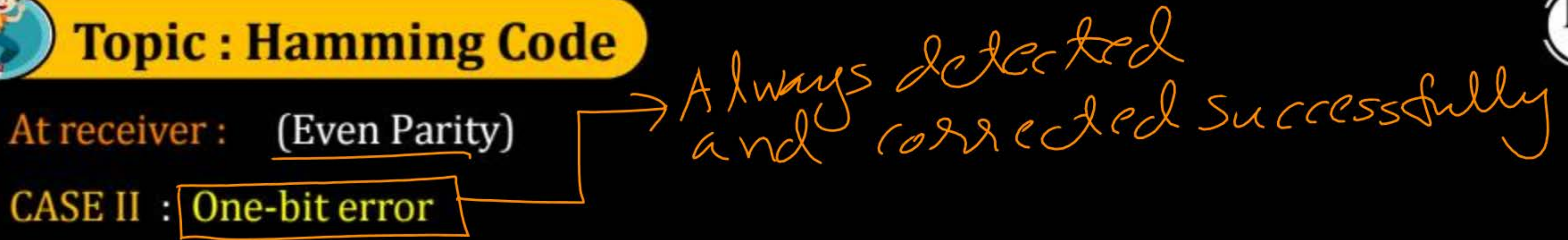
$R_0 \Rightarrow X$

$R_1 \Rightarrow X$

$R_2 \Rightarrow X$

$R_3 \Rightarrow \checkmark$

If one-bit error in data than error must be in position 7.



CASE II : One-bit error

[Receiver finds parity bits R_0, R_1 , and R_2 are unbalanced]

Error Position = 7 [R₃ R₂ R₁ R₀ = 0 1 1 1]

[Receiver successfully corrected single-bit error]



Topic : Hamming Code



At receiver : (Even Parity)

CASE III : Two-bit error

Always detected,
But not corrected;

Valid Code,
Transmitted Codeword = 101001101100

Received Codeword = 101000111100

$R_0 \Rightarrow \checkmark$
 $R_1 \Rightarrow \times$
 $R_2 \Rightarrow \checkmark$
 $R_3 \Rightarrow \checkmark$

If one-bit error
than error must
be in position 2.

<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
12	11	10	9	8	7	6	5	4	3	2	1



Topic : Hamming Code



At receiver : (Even Parity)

CASE III : Two-bit error

Receiver Concluded : “Error detected”
[Receiver finds only parity bits R_1 is unbalanced]

Error Position = 2 [$R_3 R_2 R_1 R_0 = 0 0 \underline{1} 0$]

Corrected Codeword = 1 0 1 0 0 0 1 1 1 1 0

[Receiver can not correct burst error]

→ Valid Code₂



Topic : Hamming Code



At receiver : (Even Parity)

CASE IV : Three-bit error

Transmitted Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

Received Codeword = 1 0 1 0 0 1 1 0 1 0 1 1



$R_0 \Rightarrow \checkmark$
 $R_1 \Rightarrow \checkmark$
 $R_2 \Rightarrow \checkmark$
 $R_3 \Rightarrow \checkmark$



Topic : Hamming Code



At receiver : (Even Parity)

CASE IV : Three-bit error

→ May be detected;

Receiver Concluded : “No any error detected”
[Receiver finds all parity bits are balanced]

[MCQ]

IIT-B (H.W.)

[GATE-2021][2 Mark]



#Q. Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$, where the data bits and the check bits are given in the following tables:

Data bits							
d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1
1	1	0	x	0	1	0	1

Check bits			
c_8	c_4	c_2	c_1
y	0	1	0

Which of the following choices gives the correct values of x and y?

A

x is 0 and y is 0

B

x is 0 and y is 1

C

x is 1 and y is 0

D

x is 1 and y is 1



Topic : Hamming Code



→ Number of data bits = m bits

→ Number of parity bits = r bits

→ Code length (N) = $[m + r]$ bits

→ Hamming (N, m)

hamming (N, m, d)

$d \rightarrow \min^m$ Hamming Distance



Topic : Hamming Code



→ Minimum number of parity bits required

$$\Rightarrow 2^r > (m + r)$$

$$(2^r - 1) \geq (m + r)$$

$$2^r \geq (m + r + 1)$$

Using r parity bit :-

$$\text{max}^m \text{ code length} = (2^r - 1)$$

$$\text{code length} = (m + r)$$

m	r	
1	2	$\Rightarrow (3)$
2	3	
3	3	
4	3	$\Rightarrow (7)$
5	4	
6	4	
7	4	
11	4	$\Rightarrow (15)$
12	5	
26	5	$\Rightarrow (31)$
27	6	



Topic : Hamming Code



→ Minimum code length = 3 ✓
[contains only one data bit and two parity bit]

→ Code length $\neq 2^i$ ✓
[where $i = 0, 1, 2, 3 \dots$]



Topic : Hamming Code



→ Using **r** parity bit,

Maximum code length

=

$$(2^r - 1)$$

Maximum data bits

=

$$(2^r - r - 1)$$

→ Code rate

=

$$(2^r - r - 1) / (2^r - 1)$$

$$\text{code rate} = \frac{m}{(m+r)} = \frac{\text{No. of data bits in code}}{\text{Total no. of bits in code}}$$



Topic : Hamming Code



Hamming Code (with Even Parity) and 3 data bits:

Data → Codeword
d₂d₁d₀ → d₂d₁C₂d₀C₁C₀

0 0 0	→	0 0 0 0 0 0
0 0 1	→	0 0 0 1 1 1
0 1 0	→	0 1 1 0 0 1
0 1 1	→	0 1 1 1 1 0
1 0 0	→	1 0 1 0 1 0
1 0 1	→	1 0 1 1 0 1
1 1 0	→	1 1 0 0 1 1
1 1 1	→	1 1 0 1 0 0

2^3 data

2^3 Valid Codeword

Linear code →

min^m Hamming Distance
= min^m hamming weight
of non-zero codewords

min^m H.D. ⇒ 3 (always)



Topic : Hamming Distance



if (minimum) Hamming Distance is D

then receiver can detect upto $(D-1)$ bits error

and receiver can correct upto $\text{Floor}[(D-1)/2]$ bits error

$$\left\lfloor \frac{(D-1)}{2} \right\rfloor$$



Topic : Hamming Distance



To detect up to x - bits error

-> minimum Hamming Distance should be $(x+1)$

To correct up to y - bits error

-> minimum Hamming Distance should be $(2y+1)$



Topic : Hamming Distance

$$\boxed{\min^m \text{H.D.} \geq 2}$$



→ (minimum) Hamming Distance in:

1. One-bit parity is always "2".

→ Always detect "single bit error" → No any error correction

2. Hamming Code is always "3".

→ Always correct "single bit error" ✓

→ Always detect "upto two bit error"

⇒ 3-bit error may be detected

3. Two-dimension parity is always "4".

→ Always correct "single bit error" ✓

→ Always detect "upto three bit error"

⇒ 4-bit error may be detected



Topic : One-bit parity

Linear code



One-bit parity (with even parity) and 3 data bits

Data --> Codeword
 $d_1 d_2 d_3$ --> $d_1 d_2 d_3 P$

0 0 0	-->	0 0 0 0
0 0 1	-->	0 0 1 1
0 1 0	-->	0 1 0 1
0 1 1	-->	0 1 1 0
1 0 0	-->	1 0 0 1
1 0 1	-->	1 0 1 0
1 1 0	-->	1 1 0 0
1 1 1	-->	1 1 1 1



Topic : Two-dimension (2D) Parity



Linear code
 $\min^m H.D. = 4$

Two-dimension Parity (with Even Parity) and 4 data bits :

Data	→	Codeword	Data	→	Codeword
$d_1d_2d_3d_4$	→	$d_1d_2R_1d_3d_4R_2C_1C_2P$	$d_1d_2d_3d_4$	→	$d_1d_2R_1d_3d_4R_2C_1C_2P$
0 0 0 0	→	0 0 0 0 0 0 0 0	1 0 0 0	→	1 0 1 0 0 0 1 0
0 0 0 1	→	0 0 0 0 1 1 0 1	1 0 0 1	→	1 0 1 0 1 1 1 0
0 0 1 0	→	0 0 0 1 0 1 1 0	1 0 1 0	→	1 0 1 1 0 1 0 0
0 0 1 1	→	0 0 0 1 1 0 1 1	1 0 1 1	→	1 0 1 1 1 0 0 1
0 1 0 0	→	0 1 1 0 0 0 0 1	1 1 0 0	→	1 1 0 0 0 0 1 1
0 1 0 1	→	0 1 1 0 1 1 0 0	1 1 0 1	→	1 1 0 0 1 1 1 0
0 1 1 0	→	0 1 1 1 0 1 1 1	1 1 1 0	→	1 1 0 1 0 1 0 1
0 1 1 1	→	0 1 1 1 1 0 1 0	1 1 1 1	→	1 1 0 1 1 0 0 0



Topic : Hamming Code



Hamming Code (with Even Parity) and 3 data bits :

Data	→	Codeword
$d_2d_1d_0$	→	$d_2d_1C_2d_0C_1C_0$

0 0 0	→	0 0 0 0 0 0
0 0 1	→	0 0 0 1 1 1
0 1 0	→	0 1 1 0 0 1
0 1 1	→	0 1 1 1 1 0
1 0 0	→	1 0 1 0 1 0
1 0 1	→	1 0 1 1 0 1
1 1 0	→	1 1 0 0 1 1
1 1 1	→	1 1 0 1 0 0

[H.W.]

[GATE-1992]



- #Q. Consider a 3-bit error detection and 1-bit error correction hamming code for 4-bit data. The extra parity bits required would be _____ and the 3-bit error detection is possible because the code has a minimum distance of _____ .



Topic : Optimal Packet Size



→ Suppose each link have same bandwidth

$$\underline{t_x} = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{(H+P)}{\text{Bandwidth}}$$

K = Number of Routers = 2
M = Message Size (in bytes)
H = Packet Header Size (bytes)
P = Packet Payload Size (bytes)

Packet $\Rightarrow$ IP Packet
 $\Rightarrow$ IP Datagram

$$\text{Number of packets (N)} = \frac{\text{Message Size}}{\text{Payload Size}} = \frac{M}{P}$$



Topic : Optimal Packet Size



→ Total time required to transmit one message completely from Sender to Receiver

$$\begin{aligned}\text{Total Time (T)} &= [N * \underline{t_x} + \underline{t_{p1}}] + [\underline{t_x} + \underline{t_{p2}}] + [\underline{t_x} + \underline{t_{p3}}] \\&= \underline{(N + K) * t_x} + [\underline{t_{p1}} + \underline{t_{p2}} + \underline{t_{p3}}] \\&= \underline{(N + K) * t_x} + \underline{t_p} \\&\quad [\text{Consider negligible propagation delay}] \\&= (N + K) * t_x \\T &= \frac{(N + K) * (H + P)}{\text{Bandwidth}}\end{aligned}$$



Topic : Optimal Packet Size



$$\text{Total Time (T)} = \frac{(N + K) * (H + P)}{\text{Bandwidth}}$$

→ For what value of P, total time T should be minimum :

$$\frac{dT}{dP} = 0$$

$$P = \sqrt{\frac{MH}{K}}$$

$$\text{Optimal packet size} = (H + P)$$





Topic : Optimal Packet Size



$$T = \frac{(N + K) * (H + P)}{\text{Bandwidth}}$$

$$\frac{dT}{dP} = \frac{d}{dP} \left[\frac{(N+K) * (H+P)}{\text{Bandwidth}} \right] = 0$$

$$\frac{d}{dP} \left[\left(\frac{M}{P} + K \right) * (H+P) \right] = 0$$



Topic : Optimal Packet Size



$$\frac{d}{dP} \left(\frac{M}{P} + K \right) * (H + P) = 0$$

$$\left(\frac{M}{P} + K \right) * \frac{d}{dP} (H + P) + (H + P) * \frac{d}{dP} \left(\frac{M}{P} + K \right) = 0$$

$$\left(\frac{M}{P} + K \right) * 1 + (H + P) * \left[M * \left(\frac{-1}{P^2} \right) \right] = 0$$

$$\frac{M}{P} + K - \frac{MH}{P^2} - \frac{M}{P} = 0$$

$$P = \sqrt{\frac{MH}{K}}$$

[MCQ]

IIT-B

[GATE-CS-2005] [2 Mark]



#Q. In a packet switching network, packets are routed from source to destination along a single path having two intermediate nodes. If the message size is 24 bytes and each packet contains a header of 3 bytes, then the optimum packet size is :

A 4

~~B 6~~

C 7

☒ D 9

No. of nodes/Routers (K) = 2

Message Size (M) = 24 bytes

IP Packet Header Size (H) = 3 bytes

$$P = \sqrt{\frac{MH}{K}} = \sqrt{\frac{24 \text{ byte} \times 3 \text{ byte}}{2}}$$

$$P = \sqrt{(6 \text{ byte})^2} = 6 \text{ byte}$$

P → IP Packet
Payload size (in bytes)

Optimum Payload size = 6 byte

Optimum Packet size
= (H + P) = (3 + 6) byte
= 9 bytes

Ans: D



2 mins Summary



Topic

Hamming Code ✓

Topic

Hamming Distance ✓

Topic

Optimum Packet Size ✓



THANK - YOU

